

fixed

Fixed

Revision: 211 **Last modified:** 2026-06-16T15:17:22Z **Description:** Completed workable items (with evidence references) **Authority:** Constitution §11.4.93 (workable-items DB single source of truth) **Generated-by:** scripts/docs/db_to_md.py (DB is canonical; edit via cmd/hxc-registry, not by hand)

Total completed: 777.

HXC	Type	Pri	Title
HXC-001	Task	P1	
HXC-002	Task	P1	
HXC-003	Task	P1	
HXC-004	Task	P1	
HXC-005	Task	P1	
HXC-006	Task	P1	
HXC-007	Task	P1	
HXC-008	Task	P1	
HXC-009	Task	P1	
HXC-010	Task	P1	
HXC-011	Task	P1	
HXC-012	Task	P1	
HXC-013	Task	P1	
HXC-014	Task	P1	
HXC-015	Task	P1	
HXC-1000	Bug	P0	FIX delta-ORSet strong-eventual-consistency defect (remove-before-add)
HXC-1001	Task	P0	Stand up embedded-etcd integration harness for pkg/etcd tests
HXC-1002	Feature	P0	Prove etcd Put/Get round-trip and missing-key semantics against real etcd
HXC-1003	Feature	P1	Prove etcd GetPrefix returns exactly matching multi-key set
HXC-1004	Feature	P1	Prove etcd Delete/DeletePrefix remove keys (sink-side absence)
HXC-1005	Feature	P0	Prove etcd Watch fires correct event and closes on ctx cancel
HXC-1006	Feature	P1	Prove etcd Lease/PutWithLease/RevokeLease/KeepAlive TTL behavior
HXC-1007	Feature	P0	Prove etcd distributed Lock mutual exclusion (highest-risk path)
HXC-1008	Task	P2	Add mutation-paired etcd New() default-timeout injection test

HXC	Type	Pri	Title
HXC-1009	Task	P2	Prove etcd error-wrapping on dead endpoint
HXC-1010	Feature	P0	Replace pkg/infra map-echo orchestrator with real service Boot via tests
HXC-1011	Feature	P0	Make pkg/infra Health probe real services and report false on failure
HXC-1012	Feature	P1	Make pkg/infra Logs capture real sink-side output honoring tail/since/timestamp
HXC-1013	Feature	P1	Make pkg/infra Scale create real replicas verified via runtime query
HXC-1014	Feature	P1	Make pkg/infra VMSpawn/VMSSH establish a real session end-to-end
HXC-1015	Feature	P2	Make pkg/infra VMSimulateFailure/VMSimulatePartition assert observability
HXC-1016	Feature	P0	Add EtcdLocker integration tests against real etcd (block/acquire/lease)
HXC-1017	Bug	P0	Rewrite TestMemoryLockerConcurrent to detect a broken lock without a lock
HXC-1018	Task	P2	Add explicit block-then-acquire ordering test for MemoryLocker
HXC-1019	Bug	P2	Characterize or fix MemoryLocker.Lock context-cancellation goroutine
HXC-1020	Feature	P0	Wire a real Builder behind pkg/build Service and prove image is pullable
HXC-1021	Bug	P1	Drive pkg/build StateFailed via a genuine build failure not the 'fail' sentinel
HXC-1022	Bug	P1	Make pkg/build List/Concurrent tests Start() the service and assert termination
HXC-1023	Task	P2	Replace time.Sleep polling in pkg/build with completion sync and add some mutation tests
HXC-1024	Bug	P2	Enforce or document pkg/build content-addressable cache digest integrity
HXC-1025	Task	P2	Cover pkg/build cancel-mid-flight branch sets StateCancelled
HXC-1026	Feature	P0	Add real mTLS handshake test for pkg/security (valid client succeeds, non-rejected)
HXC-1027	Bug	P1	Assert MinVersion==TLS13 on the client TLS path in pkg/security
HXC-1028	Feature	P0	Locate/build production Vault client and add dev-Vault integration test (CI)
HXC-1029	Task	P3	Replace pkg/security non-nil-only TestNewTLSConfigBuilder with meaningful
HXC-1030	Bug	P1	Verify pkg/hxcregistry item_history audit-sink rows after Create/Update/Delete
HXC-1031	Bug	P2	Assert exact and stable ComputeHeadingHash value in pkg/hxcregistry
HXC-1032	Task	P2	Add pkg/hxcregistry failure-path tests (dup PK, missing-id Get/Update)
HXC-1033	Bug	P2	Make pkg/hxcregistry concurrent test do concurrent writes (or rename)
HXC-1034	Task	P3	Prove pkg/hxcregistry timestamp round-trip and DB CHECK-constraint
HXC-1035	Bug	P0	Wire pkg/wasm WasmPlugin.Execute to real WASM output and prove it
HXC-1036	Task	P1	Add pkg/wasm negative tests for missing execute export and corrupt module
HXC-1037	Bug	P2	Prove pkg/wasm init/shutdown exports actually run via memory side-effects
HXC-1038	Task	P3	Cover pkg/wasm Host.Call ErrUnexpectedResult branch
HXC-1039	Bug	P1	Replace self-defeating pkg/semaphore TestSemaphore_ReleaseTooMany
HXC-1040	Feature	P1	Add pkg/semaphore M>>N concurrency invariant test under -race
HXC-1041	Task	P3	Add pkg/semaphore New(0) barrier and exact-capacity boundary tests
HXC-1042	Bug	P1	Convert pkg/storage error tests to errors.Is on exported sentinels
HXC-1043	Bug	P2	Reconcile and test pkg/storage FileStore vs MemoryStore copy semantics
HXC-1044	Task	P2	Add pkg/storage overwrite, missing-delete, and List edge tests for both s
HXC-1045	Feature	P1	Add pkg/storage -race concurrent Put/Get/Delete test to prove RWMutex
HXC-1046	Bug	P1	Gate -race in CI for pkg/metrics concurrency mutation tests

HXC	Type	Pri	Title
HXC-1047	Bug	P2	Complete pkg/metrics Prometheus exposition assertions (+Inf, _sum, or
HXC-1048	Task	P2	Add pkg/metrics duplicate-registration and histogram edge tests
HXC-1049	Task	P3	Add pkg/metrics /metrics integration test via httptest.Server
HXC-1050	Bug	P1	Test pkg/grpcutil LoggingStreamInterceptor recv/send counters
HXC-1051	Bug	P2	Strengthen pkg/grpcutil log assertions to dynamic behavior-dependent fi
HXC-1052	Task	P2	Cover pkg/grpcutil status.FromError code extraction for handler errors
HXC-1053	Bug	P1	Add pkg/context TestDetach_PreservesValues for the value-retention gu
HXC-1054	Bug	P2	Strengthen pkg/context WithTimeout deadline and expiry assertions
HXC-1055	Bug	P1	Give pkg/workerpool TestPool_NilWorkSafe_Mutation a real sink
HXC-1056	Feature	P1	Add pkg/workerpool fan-in count==N, post-Stop Submit, panic-recover, tests
HXC-1057	Bug	P1	Make pkg/errors concurrency test honest via locked GetFields path + -ra
HXC-1058	Task	P2	Pin pkg/errors Code enum exact values, distinctness, and WithFields nil
HXC-1059	Bug	P2	Replace tautological pkg/backoff cap-mutation test and pin Default()
HXC-1060	Bug	P2	Assert pkg/classads parser AST Op per operator and paren precedence
HXC-1061	Task	P3	Add pkg/classads eval failure-path and regexp non-match tests
HXC-1062	Bug	P2	Add pkg/crypto SHA-256 known-answer vectors and sentinel errors.Is c
HXC-1063	Task	P3	Add pkg/crypto AES-128/192, wrong-key decrypt, short-ciphertext, PBI
HXC-1064	Bug	P1	Add pkg/log prefix-field proof catching the SetLevel prefix-loss bug
HXC-1065	Task	P3	Strengthen pkg/log Warn test, add Debug-suppression, remove dead cap scaffolding
HXC-1066	Task	P3	Add pkg/lru boundary, cold-miss, and struct-value round-trip tests
HXC-1067	Feature	P2	Add pkg/pubsub -race concurrency test and explicit drop-boundary asse
HXC-1068	Bug	P2	Prove pkg/ratelimit bucket starts full and add Limiter conformance + -ra
HXC-1069	Bug	P1	Bring pkg/retry DoWithResult to behavioral parity with Do
HXC-1070	Task	P2	Add pkg/retry transient-recovery, jitter-bound, and computeDelay table
HXC-1071	Task	P3	Strengthen pkg/serde MustMarshal exact-content and add Unmarshal fa
HXC-1072	Bug	P2	Pin pkg/validator error contract via errors.Is and remove/wire unused se
HXC-1073	Task	P3	Add pkg/validator uint min/max, IsValidID empty, oneof/email boundar
HXC-1074	Task	P0	Establish fleet-wide -race CI gate across concurrency-sensitive packages
HXC-1075	Feature	P0	Implement NATS/JetStream client wrapper (pkg/events)
HXC-1076	Feature	P0	Implement etcd client wrapper (pkg/etcd)
HXC-1077	Feature	P0	Implement etcd discovery backend (pkg/discovery)
HXC-1078	Feature	P0	Implement real Kafka producer/consumer (pkg/pubsub)
HXC-1079	Feature	P0	Implement cgroups v2 + /proc resource reader (pkg/resources)
HXC-1080	Feature	P0	Implement Omega-model scheduler core (pkg/scheduler, internal/schedu
HXC-1081	Feature	P0	Implement optimistic-concurrency versioning for scheduler pool
HXC-1082	Feature	P1	Implement HTCondor ClassAds matching in scheduler (pkg/classads)
HXC-1083	Feature	P1	Implement gang scheduling and preemption in scheduler
HXC-1084	Feature	P0	Upgrade tmux backend to control-mode (-CC) with streaming I/O (pkg/

HXC	Type	Pri	Title
HXC-1085	Feature	P0	Implement CRDT session state and lifecycle (create/attach/migrate)
HXC-1086	Feature	P0	Implement real Node gRPC handlers serving helixv1 Node API (internal/node)
HXC-1087	Feature	P0	Make node registry etcd-backed and cross-process shared (internal/node)
HXC-1088	Feature	P0	Implement SWIM gossip membership and failure detection (pkg/swim)
HXC-1089	Feature	P0	Implement WireGuard mesh config-gen and key rotation (pkg/wireguard)
HXC-1090	Feature	P1	Implement WireGuard NAT traversal / STUN hole-punch (pkg/wireguard)
HXC-1091	Feature	P0	Wire API gateway reverse proxy with prefix routing and /health (internal/gateway)
HXC-1092	Feature	P0	Add JWT/RBAC auth middleware to API gateway (internal/gateway)
HXC-1093	Feature	P0	Implement leader election with fencing tokens (pkg/leader)
HXC-1094	Feature	P0	Implement distributed lock primitives over etcd (pkg/lock)
HXC-1095	Feature	P1	Implement Postgres-backed internal artifact registry (pkg/hxcregistry)
HXC-1096	Feature	P1	Implement GPU resource probe with /proc parser (internal/gpu)
HXC-1097	Feature	P1	Define vendor-agnostic GPUBackend interface and registry (internal/gpu)
HXC-1098	Feature	P2	Implement GPU sharing modes (MPS, time-slice, MIG, exclusive)
HXC-1099	Feature	P1	Implement Build Service orchestrator and worker pool (internal/build)
HXC-1101	Feature	P0	Implement e2ee transport record protocol (security/pkg/e2ee)
HXC-1102	Feature	P1	Implement software-rooted attestation (security/pkg/attestation)
HXC-1103	Feature	P0	Implement RBAC scopes and identity bindings (internal/security)
HXC-1104	Feature	P1	Implement SPIFFE/SPIRE SVID issuance and mTLS identity (internal/security)
HXC-1105	Feature	P0	Implement htmux CLI raw-mode terminal client (cmd/htmux)
HXC-1106	Feature	P0	Implement session I/O PTY-over-WebSocket forwarding (cmd/helix-session)
HXC-1107	Feature	P0	Implement WebSocket envelope and MessagePack framing (pkg/websocket)
HXC-1108	Feature	P1	Implement HelixQA live-service Challenge runner (cmd/helix-test)
HXC-1109	Feature	P1	Implement deterministic-sim, chaos, and device-sim test harness (cmd/helix-test)
HXC-1110	Task	P1	Wire Prometheus /metrics endpoint uniformly across all services (cmd/helix-test)
HXC-1111	Task	P1	Wire Jaeger/OTel trace-ID propagation across services (pkg/tracing)
HXC-1112	Task	P1	Wire pkg/health gRPC health protocol on every service (pkg/health, cmd/helix-test)
HXC-1113	Bug	P2	Remove dangling pkg/gpuattest placeholder and reuse security/pkg/gpuattest
HXC-1114	Feature	P0	Provision PostgreSQL primary schema with 15 tables, indexes, and triggers
HXC-1115	Task	P0	Define etcd key namespace and key-builder constants (pkg/etcd)
HXC-1116	Feature	P1	Define Redis cache, routing, and rate-limit key structure (pkg/storage)
HXC-1117	Task	P0	Define and generate proto service stubs (node/session/scheduler/health/alerts)
HXC-1118	Feature	P1	Implement REST API surface on API Gateway with OpenAPI 3.0
HXC-1119	Feature	P1	Implement Avro event schemas and schema-validated routing (Event Bus)
HXC-1120	Task	P1	Configure Kafka topics with partitions/replication/retention
HXC-1121	Task	P0	Configure NATS JetStream streams (HELIX_NODES/SESSIONS/SCHEDULING/HEALTH/ALERTS)
HXC-1122	Feature	P2	Implement Cap'n Proto zero-copy serialization for control-plane (pkg/session)
HXC-1123	Feature	P2	Implement FlatBuffers serialization for GPU compute payloads
HXC-1124	Feature	P2	Implement ZeroMQ message patterns for internal data planes

HXC	Type	Pri	Title
HXC-1125	Feature	P1	Implement Setup Wizard single-command node onboarding (cmd, BAS)
HXC-1126	Feature	P2	Implement LLM Brain advisory engine with LLMsVerifier gate (internal)
HXC-1127	Feature	P1	Implement OPA/WASM Policy Engine with HelixConstitution enforcement
HXC-1130	Feature	P2	Implement Zellij and GNU screen session backends
HXC-1131	Feature	P2	Implement Backup Service (etcd snapshots, WAL archival, Ceph checkpoints)
HXC-1132	Feature	P1	Implement Metrics Collector node scrape endpoint and GPU metrics aggregation
HXC-1133	Feature	P1	Implement token-bucket rate limiter for API gateway (pkg/ratelimit)
HXC-1134	Feature	P2	Implement Vault secret injection and rotation for all services
HXC-1135	Feature	P0	Implement two-node cluster formation end-to-end (exit gate)
HXC-1136	Feature	P0	Implement session-create end-to-end path (htmux new exit gate)
HXC-1137	Task	P1	Benchmark job scheduling decision latency under 100ms (exit gate)
HXC-1138	Task	P1	Enforce >60% pkg/ line coverage and paired mutation tests (Constitution)
HXC-1139	Research	P2	Author TLA+ specifications for consensus and scheduling concurrency
HXC-1141	Feature	P2	Implement interactive-mode AI CLI agent resource provisioning
HXC-1142	Feature	P2	Implement hybrid mode switching and batch/interactive resource sharing
HXC-1144	Feature	P2	Implement WireGuard mesh network policies (segmentation)
HXC-1145	Docs	P2	Author MVP architecture and component-specification documentation and
HXC-1146	Docs	P0	Document node-provisioning boundary: project does NOT implement jail
HXC-1147	Feature	P1	Implement internal/console/linux_boot.go BootCoordinator state machine
HXC-1148	Feature	P1	Replace pkg/wireguard NAT discovery with RFC 5389 STUN client
HXC-1149	Task	P3	Return typed ErrUnsupported for UPnP/NAT-PMP in pkg/wireguard
HXC-1150	Feature	P1	Add WireGuard key-rotation grace/overlap window with ActiveKeys()
HXC-1151	Task	P1	Add pkg/discovery etcd-backed integration test against a real etcd
HXC-1152	Feature	P1	Implement pkg/session container-checkpoint migration strategy (CRDT)
HXC-1156	Feature	P1	Implement thermal-aware scheduler plugin (throttle on overheat)
HXC-1157	Feature	P1	Implement semi-trusted node output verification (redundant-compute/ch
HXC-1158	Task	P2	Harden device hardware-identification in internal/console/detector.go (g
HXC-1160	Feature	P2	Improve tmux session backend to full PTY wrap on Attach
HXC-1162	Task	P2	E2E: multi-node session CRDT convergence
HXC-1164	Docs	P2	Author phase_02 architecture diagram (operator console + mesh + node
HXC-1165	Task	P2	Ensure api/v1/node.proto carries trust_level + thermal fields consumed e
HXC-1167	Task	P0	Build ARM64 cross-compilation toolchain for Helix agent
HXC-1175	Feature	P0	Implement charging-gated PowerGater scheduling guard
HXC-1178	Feature	P0	Implement MQTT client for edge work dispatch and status
HXC-1184	Feature	P0	Implement edge node registration for all tiers (T3-T8)
HXC-1185	Feature	P0	Implement edge heartbeat with battery/thermal/network telemetry
HXC-1186	Feature	P0	Implement edge protocol gateway (MQTT/QUIC/WebSocket)
HXC-1187	Feature	P0	Define EdgeWorkUnit and EdgeWorkResult protobuf schemas

HXC	Type	Pri	Title
HXC-1188	Feature	P0	Enforce work-unit resource limits (duration/memory/CPU) on edge device
HXC-1189	Feature	P0	Implement EdgeAwarePlugin scheduler Filter stage
HXC-1190	Feature	P1	Implement EdgeAwarePlugin scheduler Score stage
HXC-1191	Feature	P1	Implement declarative per-tier ScheduleRule engine
HXC-1193	Feature	P1	Implement per-tier workload quantization selection
HXC-1195	Feature	P0	Implement edge trust-level model (STANDARD/SEMI/EDGE_DONOR)
HXC-1196	Feature	P0	Enforce workload restriction matrix by trust level
HXC-1197	Feature	P0	Implement edge output verification (LLMsVerifier/redundant/checksum)
HXC-1198	Feature	P1	Implement offline sync protocol with delta compression
HXC-1199	Feature	P1	Implement edge sensor-fusion framework and stream workload type
HXC-1201	Feature	P2	Implement low-bandwidth protocol optimizations (CoAP option)
HXC-1202	Research	P3	Implement WebRTC P2P data-channel transport for edge devices
HXC-1207	Task	P0	Implement edge device chaos test suite
HXC-1210	Bug	P0	Wire EtcdBackend into the node agent for cluster-wide discovery
HXC-1211	Bug	P0	Make internal/scheduler.StreamJobEvents a real event stream
HXC-1212	Feature	P0	Connect pkg/classads to scheduler Filter/Score stages
HXC-1213	Feature	P0	Add auth + RBAC enforcement to the gateway
HXC-1214	Feature	P1	Add S3-compatible (minio) backend to pkg/storage
HXC-1215	Feature	P1	Add OTLP/Jaeger export shim for pkg/tracing
HXC-1216	Feature	P2	Implement true embedded Raft consensus (pkg/raft)
HXC-1220	Task	P2	Replace custom policy engine with OPA/Rego
HXC-1229	Task	P0	Build device profile registry with versioned YAML schema for T1-T8
HXC-1230	Feature	P1	Implement automated tier detection validating host KVM/ARM64/binfmt
HXC-1231	Feature	P1	Implement device provisioning lifecycle abstraction (Provisioner/Instantiator)
HXC-1233	Feature	P1	Implement qcow2 copy-on-write overlay management with 10-deep chain
HXC-1236	Feature	P0	Implement INetwork/HelixNetwork trait with production and simulation
HXC-1237	Feature	P0	Wire DST sim transport seam into pkg/swim membership (prod+sim parallel)
HXC-1238	Feature	P1	Implement BUGGIFY macro framework with ~25% deterministic fire rate
HXC-1239	Feature	P1	Implement 10:1 DST virtual-time compression by fast-forwarding idle p
HXC-1240	Feature	P1	Achieve 1,000+ simulated nodes in a single DST process
HXC-1241	Feature	P1	Author DST consensus+gossip workloads using SETUP→EXECUTION→CHECK→METRICS pattern
HXC-1243	Feature	P0	Implement 8 network fault injectors (latency, loss, corruption, reorder, d partition, DNS, TCP reset)
HXC-1244	Feature	P0	Implement 8 node fault injectors (VM crash/restart/pause, CPU/mem/disk kill, graceful shutdown)
HXC-1245	Feature	P1	Implement 3 time fault injectors (clock skew, clock freeze, monotonic d
HXC-1247	Feature	P1	Implement pure-Go in-sim chaos faults toward 25+ (ClockSkew, DiskFi MessageReorder, Byzantine, etc.)
HXC-1248	Feature	P0	Build YAML chaos Scenario Engine with phases, blast radius and abort-

HXC	Type	Pri	Title
HXC-1249	Feature	P0	Implement emergency-stop and auto-recovery with <=2s halt latency
HXC-1254	Feature	P1	Implement TestRunner with parallel suite execution and result collection
HXC-1255	Feature	P1	Implement session test state machine (IDLE→SETUP→RUNNING→CHAOS_INJECT→VERIFY→RECO
HXC-1258	Feature	P1	Implement MetricsCollector exporting 15+ chaos Prometheus metric ser OpenTelemetry tracing
HXC-1259	Feature	P1	Implement HelixQA automatic challenge generation from test outcomes
HXC-1260	Feature	P1	Implement metrics validation against baseline KPI table with severity ga
HXC-1261	Feature	P1	Implement Welch's t-test statistical regression detector for HelixQA
HXC-1264	Task	P1	Define WIT interfaces for device-simulator, workload-generator, fault-in exporter
HXC-1265	Feature	P1	Implement capability-based plugin sandbox with resource limits and auc
HXC-1267	Feature	P0	Implement cmd/helix-test CLI with dst/chaos/device/snapshot subcomm
HXC-1268	Feature	P1	Build cmd/helix-snapshot standalone CLI reusing pkg/testing/snapshot
HXC-1278	Research	P3	Add property-based + coverage-guided fuzzing for cluster operations
HXC-1290	Feature	P0	Implement pkg/device universal capability Descriptor with 15-tier + trus
HXC-1291	Feature	P0	Implement device.Probe(ctx) arch/CPU/mem auto-detection with overri
HXC-1292	Feature	P0	Implement device.AssignTier decision tree per §50 taxonomy (riscv64-> fw->T1, GPU+battery->T9)
HXC-1293	Feature	P0	Extend api/v1/node.proto with tier, trust_level, compute_class, arch field
HXC-1294	Feature	P0	Create api/v1/resources.proto with npu_tops, fpga_logic_elements, tflop
HXC-1295	Feature	P0	Extend api/v1/scheduler.proto with tier constraints and power_budget
HXC-1296	Feature	P0	Implement device.Descriptor <-> pb.Node round-trip mapping helpers
HXC-1297	Feature	P0	Implement pkg/scheduler tier-aware + power-aware Plugin (tier_power,
HXC-1298	Feature	P0	Implement power-aware T9 HANDHELD scheduling policy (battery/th active gating)
HXC-1299	Feature	P0	Implement pkg/discovery 15-tier registry + SelectByTier(tier, minTrust)
HXC-1300	Feature	P0	Extend pkg/resources to probe NPU TOPS, FPGA logic elements, GPU
HXC-1301	Feature	P1	Implement pkg/cloudspot PreemptionWatcher with injectable SignalSou hooks
HXC-1302	Feature	P1	Extend pkg/wireguard with preemption-safe teardown / spot-drain hook
HXC-1303	Feature	P1	Implement pkg/inference provider-agnostic Backend interface + capabil
HXC-1304	Feature	P1	Implement pkg/inference LocalBackend wrapping internal/llm manager
HXC-1305	Feature	P1	Implement device discovery engine: GPU detection (Vulkan/CUDA/RO strategies)
HXC-1306	Feature	P1	Implement device discovery engine: NPU detection (Rockchip/NVIDIA Apple)
HXC-1307	Feature	P2	Implement device discovery engine: FPGA detection (Xilinx/Intel/Lattic core)
HXC-1308	Feature	P2	Implement micro-benchmark producing normalized per-compute-class
HXC-1309	Feature	P1	Implement capability manifest generation + control-plane negotiation
HXC-1310	Feature	P0	

HXC	Type	Pri	Title
			Implement per-tier security model enforcement (sandbox + data/network level)
HXC-1311	Docs	P1	Author complete YAML tier definitions (T1-T15) with min_requirements
HXC-1328	Feature	P1	Implement cloud-spot AWS/Azure/GCP IMDS interruption pollers (adapted)
HXC-1331	Docs	P2	Author 64-device master taxonomy table as machine-readable device catalog
HXC-1334	Feature	P0	Implement pkg/hlc Hybrid Logical Clock with drift clamp and total order
HXC-1335	Feature	P0	Implement pkg/crdt G-Counter with per-replica max-merge convergence
HXC-1336	Feature	P0	Implement pkg/crdt LWW-Register with HLC last-writer-wins tiebreak
HXC-1337	Feature	P0	Implement pkg/crdt OR-Set with unique-tag add/remove semantics
HXC-1338	Feature	P1	Implement pkg/crdt/merkle anti-entropy tree with divergent-key Diff
HXC-1339	Feature	P1	Implement Vector Clock causality type for Tier-2 operational state
HXC-1340	Feature	P0	Implement pkg/federation Cell model and lifecycle state machine
HXC-1341	Feature	P0	Implement thread-safe Cell Registry with add/remove/list/snapshot
HXC-1342	Feature	P1	Implement pkg/federation federated API aggregation with partial failure
HXC-1343	Feature	P1	Implement Phi-accrual failure detector in pkg/swim
HXC-1344	Feature	P0	Implement pkg/swim/hierarchical two-tier LAN+WAN delegate gossip
HXC-1345	Feature	P1	Implement gateway-relayed cross-cell suspicion propagation
HXC-1346	Feature	P1	Implement bootstrap/rendezvous strategies (static, DNS-SRV, DHT, cloud)
HXC-1347	Feature	P1	Implement mDNS/DNS-SD local cell discovery advertiser and browser
HXC-1348	Feature	P0	Implement pkg/nat traversal STUN client and NAT-type classification
HXC-1349	Feature	P0	Implement ICE candidate gathering, prioritization and connectivity check
HXC-1350	Feature	P1	Implement TURN relay allocation and fallback chain
HXC-1351	Feature	P1	Implement QUIC fallback transport with 0-RTT and connection migration
HXC-1352	Feature	P0	Implement internal/cell WireGuard inter-cell mesh manager
HXC-1353	Feature	P1	Implement WireGuard zero-downtime key rotation with overlap window
HXC-1356	Feature	P2	Integrate libp2p DHT discovery, GossipSub and circuit relay
HXC-1357	Feature	P1	Implement pkg/spiffe/federation trust-bundle exchange across cells
HXC-1358	Feature	P2	Implement OPA/Gatekeeper cross-cluster federated-trust admission
HXC-1359	Feature	P2	Implement OPA data-residency / data-sovereignty admission policy
HXC-1361	Feature	P1	Implement double-encryption (WireGuard L3 + mTLS L7) data path
HXC-1363	Feature	P1	Implement federated service discovery (cell-local + global registry)
HXC-1364	Feature	P1	Implement internal/federation Karmada PropagationPolicy engine
HXC-1365	Feature	P1	Extend pkg/scheduler with latency-aware spot/preemptible scoring
HXC-1367	Feature	P1	Implement pkg/gitops ArgoCD ApplicationSet federation client
HXC-1368	Feature	P2	Implement CRDT-based config sync for cell-local overrides
HXC-1371	Task	P1	Enforce etcd-stays-within-cell static-analysis lint gate
HXC-1372	Feature	P2	Implement per-cell etcd Raft WAN-safe tuning profiles
HXC-1373	Feature	P1	Implement split-brain prevention and partition-classification logic
HXC-1374	Feature	P2	Implement internal/chaos CE-01..CE-12 chaos experiment suite
HXC-1375	Feature	P2	Implement Turmoil-style deterministic multi-cluster simulation harness

HXC	Type	Pri	Title
HXC-1377	Docs	P2	Author Phase 6 FMEA catalog with detection and recovery runbooks
HXC-1380	Feature	P1	Implement split-brain detection PromQL alerts with runbook links
HXC-1381	Feature	P3	Build Grafana global-health cell-status-grid dashboard
HXC-1383	Feature	P2	Implement five federation topology patterns and block-binding modes
HXC-1386	Task	P1	Verify sub-phase dependency gates 6a->6b->6c->6d soak conditions
HXC-1387	Feature	P0	Implement Multi-Raft per-shard consensus manager (G-01, P0)
HXC-1388	Feature	P0	Implement heartbeat coalescer for Multi-Raft network overhead control
HXC-1389	Feature	P1	Implement leaseholder fast-path local reads in Multi-Raft
HXC-1390	Feature	P0	Implement MVCC revision store with B-tree index and time-travel (G-02, P0)
HXC-1391	Feature	P1	Implement persistent Watch Manager with synced/unsynced groups (G-03, P1)
HXC-1392	Feature	P0	Implement SLURM-style backfill scheduler over availability timeline (G-04, P0)
HXC-1393	Feature	P1	Implement multifactor priority queue for scheduler (age+fairshare+size+...)
HXC-1394	Feature	P0	Implement CRC16 16,384 hash-slot session router (G-03, P0)
HXC-1395	Feature	P0	Implement MOVED/ASK redirection and ASKING for in-flight slot migration
HXC-1396	Feature	P2	Implement Atomic Slot Migration controller for live session migration (G-04, P2)
HXC-1397	Feature	P2	Add config-epoch conflict resolution for simultaneous failovers
HXC-1398	Feature	P1	Add Startup health-probe tier with GPU grace period (G-04, P1)
HXC-1399	Feature	P1	Implement SWIM majority PFAIL->FAIL two-phase failure confirmation
HXC-1400	Feature	P0	Implement largest-subcluster-wins voting quorum (G-19, P0)
HXC-1401	Feature	P1	Implement STONITH fencing agents (IPMI/cloud/shared-disk) (G-19, P1)
HXC-1402	Feature	P2	Implement Pacemaker-style four-type constraint engine (G-20, P2)
HXC-1403	Feature	P2	Implement SCAN stable virtual-IP/DNS client endpoint (G-21, P2)
HXC-1404	Feature	P1	Implement N+K failover capacity admission control (G-22, P1)
HXC-1405	Feature	P1	Implement gang scheduling all-or-nothing GPU reservation (G-18, P1)
HXC-1406	Feature	P2	Implement topology-aware NUMA/NVLink GPU placement scoring (G-19, P2)
HXC-1407	Feature	P1	Implement device-plugin / GRES fingerprinting framework (G-11/G-13, P1)
HXC-1408	Feature	P1	Implement BOINC-style redundant-execution trust scorer (G-09, P1)
HXC-1409	Feature	P2	Implement standalone delta-state CRDT package (G-counter/PN-counter map) (G-01 part, P2)
HXC-1410	Feature	P1	Implement Cassandra 3-layer repair: hinted handoff + read repair + Merge (G-01 part, P1)
HXC-1411	Feature	P2	Implement Informer cache helixcache.Watcher (list-watch local cache) (G-06, P2)
HXC-1412	Feature	P2	Implement rate-limited work queue with exponential backoff (G-06, P2)
HXC-1413	Feature	P2	Implement API Priority & Fairness FlowSchema->PriorityLevel->Queue
HXC-1415	Feature	P2	Implement idempotent producer (PID + sequence) for exactly-once messaging
HXC-1416	Feature	P2	Implement cooperative incremental rebalancing for consumers/members
HXC-1417	Feature	P3	Implement embedded KRaft-style Raft quorum to remove external ZooKeeper
HXC-1418	Feature	P2	Implement tiered cache hot(memory)/warm(NVMe)/cold(SSD) data tiering
HXC-1419	Feature	P0	Build deterministic simulation testing (DST) framework on Turmoil (G-15, P0)
HXC-1420	Feature	P0	Implement BUGGIFY chaos macros at 25% deterministic fire rate (G-15, P0)

HXC	Type	Pri	Title
HXC-1421	Feature	P1	Integrate Porcupine linearizability checking in CI under fault injection (
HXC-1422	Feature	P2	Build nightly chaos pipeline (pod kill, partition, disk stall, clock skew) (
HXC-1424	Docs	P2	Produce hardened HelixCluster architecture diagram and component ma
HXC-1425	Docs	P1	Produce Phase 7 phase-by-phase 23-gap matrix as tracked deliverable le
HXC-1426	Feature	P1	Implement pkg/chutes Chutes OpenAI-compatible API Client (non-strea
HXC-1427	Feature	P1	Implement Chutes API client fallback-model chain on retrieable failure
HXC-1428	Feature	P1	Implement SSE streaming decoder for Chutes chat completions
HXC-1429	Feature	P1	Implement Chutes API model-list and user-account/balance queries
HXC-1430	Feature	P1	Implement model-router default-model resolution by strategy (latency/th
HXC-1431	Feature	P1	Wire E2EE inference envelope between pkg/chutes and security/pkg/e2e
HXC-1435	Feature	P0	Implement GraValVerifier.BatchVerify concurrent attestation with pass-r
HXC-1436	Feature	P0	Implement GraVal VRAM-threshold (95%) gate in attestation
HXC-1439	Feature	P0	Define ChutesMinerConfig and ValidatorConfig data structures
HXC-1446	Feature	P0	Implement custom HelixGepetto dual-resource arbitration strategy
HXC-1447	Feature	P0	Implement dual-workload capacity reservation in internal/gpu
HXC-1448	Feature	P1	Implement GraVal attestation hook in internal/gpu Manager
HXC-1449	Feature	P2	Implement MIG profile management in internal/gpu
HXC-1454	Feature	P1	Implement MarketplaceAdapter interface and Chutes adapter
HXC-1455	Feature	P1	Implement UnifiedManager.RouteWorkload concurrent pricing + compo
HXC-1456	Feature	P2	Implement RevenueOptimizer.OptimizeAllocation greedy GPU-to-mark
HXC-1458	Feature	P2	Implement Akash marketplace adapter
HXC-1460	Feature	P2	Implement pkg/economics RewardDistributor multi-token distribution
HXC-1461	Feature	P2	Implement RewardDistributor.GetParticipantROI and break-even trackin
HXC-1469	Feature	P1	Wire internal/gateway to route internal AI requests through Chutes API c
HXC-1470	Feature	P1	Replace internal/llm stub Inference with real model router (latency/throu
HXC-1473	Feature	P2	Enable AWQ 4-bit quantization as default model format
HXC-1476	Feature	P2	Implement hybrid PQC TLS (X25519 + ML-KEM-768) node-to-node tr
HXC-1480	Feature	P2	Implement carbon-aware scheduler with per-job energy metering
HXC-1481	Feature	P2	Implement EU AI Act compliance documentation pipeline
HXC-1482	Feature	P2	Implement export-control tier verification (country-code KYC) at node c
HXC-1483	Feature	P1	Extend pkg/metrics + Prometheus/Grafana with TAO/GraVal/throughpu
HXC-1484	Feature	P1	Extend pkg/health with miner-api and GraVal DaemonSet health checks
HXC-1493	Feature	P0	Implement GPUTier type with priority ordering (Local..Decentralized)
HXC-1494	Feature	P0	Implement PoolManager registry with tier-ordered Allocate walk
HXC-1495	Feature	P0	Implement WorkloadSpec/GPUDevice/GPUAllocation/PoolStatus data
HXC-1496	Feature	P0	Implement candidate filtering (model/VRAM/cost/label selector)
HXC-1497	Feature	P1	Implement global MaxCostPerHour budget cap on allocations

HXC	Type	Pri	Title
HXC-1498	Feature	P0	Implement PriorityScheduler (tier then least-load then cost)
HXC-1499	Feature	P1	Implement CostAwareScheduler selecting cheapest GPUs meeting SLA
HXC-1500	Feature	P1	Implement LatencyAwareScheduler for inference routing
HXC-1501	Feature	P1	Implement HealthMonitor with 30s checks and auto-failover trigger
HXC-1502	Feature	P0	Implement internal/costbroker ComputeBroker weighted scorer
HXC-1503	Feature	P1	Implement ComputeBroker 60s re-scoring loop with price-spike guard
HXC-1504	Feature	P0	Implement pkg/burst BurstController MONITOR->SPILL->RECOVER
HXC-1505	Feature	P1	Implement BurstController utilization RingBuffer moving average
HXC-1506	Feature	P1	Implement BurstController activateBurst capacity estimation + allocation
HXC-1507	Feature	P1	Implement pkg/burst CostRouter per-workload-type provider scoring
HXC-1508	Feature	P1	Implement 5-tier fallback chain Chutes->io.net->RunPod->AWS
HXC-1509	Feature	P2	Implement QoS tiers (real-time, interactive, batch, best-effort)
HXC-1510	Feature	P0	Implement pkg/provider/chutes ChutesProvider OpenAI-compatible ada
HXC-1511	Feature	P1	Implement Chutes retryWithFallback across fallback models on 429
HXC-1512	Feature	P2	Implement Chutes GetBalance USD balance monitor
HXC-1513	Feature	P2	Implement Chutes streaming (SSE) chat-completion variant
HXC-1514	Feature	P1	Implement pkg/provider/ionet IONetProvider Ray cluster adapter
HXC-1515	Feature	P1	Implement pkg/provider/runpod RunPodProvider serverless + warm pool
HXC-1516	Feature	P2	Implement pkg/provider/aws AWSProvider EC2 Spot adapter
HXC-1519	Feature	P0	Implement pkg/local LocalGPURegistrar with TCO effective-cost
HXC-1523	Feature	P1	Implement pkg/e2ee GraValVerifier provider admission gate
HXC-1527	Feature	P1	Define gpu_proxy.proto gRPC GPUProvider kernel-dispatch service
HXC-1529	Feature	P1	Implement workload-suitability classifier (HPC local vs inference remote)
HXC-1530	Feature	P0	Implement cmd/gpu-pool-manager HTTP+gRPC front-end binary
HXC-1531	Feature	P0	Implement cmd/burst-controller utilization-driven binary
HXC-1532	Feature	P0	Implement cmd/e2ee-proxy transparent ML-KEM-768 proxy binary
HXC-1533	Task	P1	Add ProviderAdapter registration hooks to pkg/gpu
HXC-1534	Feature	P1	Add GPURTier-aware filter predicate to pkg/scheduler
HXC-1535	Feature	P1	Add GPU-tier utilization/cost/provider-health metrics to pkg/metrics
HXC-1538	Feature	P2	Implement @helix.task Go decorator/SDK with lifecycle hooks
HXC-1539	Feature	P2	Implement intelligent ModelRouter (latency/throughput/cost/tee/balance)
HXC-1544	Research	P3	Implement predictive scaling forecast to pre-warm before peaks
HXC-1547	Feature	P2	Implement TCO calculator for hybrid own+burst economic model
HXC-1548	Feature	P1	Implement CostTracker monthly cost report vs AWS on-demand
HXC-1551	Task	P1	Build Prometheus/Grafana Phase 8B tier-utilization + cost dashboards
HXC-1556	Task	P1	Benchmark ML-KEM-768 E2EE handshake latency (<1ms target)
HXC-1561	Feature	P0	Add ChaCha20-Poly1305 AEAD cipher variant to pkg/security/e2ee
HXC-1563	Feature	P1	Implement bidirectional per-request response keypair in pkg/security/e2
HXC-1564	Feature	P2	Implement gzip-before-encrypt payload compression in pkg/security/e2

HXC	Type	Pri	Title
HXC-1565	Feature	P1	Implement streaming SSE E2EE decryption (e2e_init + per-chunk) in pkg/gpuattest
HXC-1567	Feature	P1	Maintain E2EE length-prefixed framed transport over io.ReadWriter
HXC-1568	Feature	P0	Implement pkg/gpuattest device-info challenge/response + fingerprint
HXC-1569	Feature	P0	Implement seeded matmul proof-of-GPU-work (PoVW) in pkg/gpuattest
HXC-1570	Feature	P0	Implement O(1) spot-check verification in pkg/gpuattest
HXC-1571	Feature	P0	Implement device-sealed encrypt/decrypt (SealForDevice/OpenFromDevice) in pkg/gpuattest
HXC-1572	Feature	P2	Implement filesystem residency challenge in pkg/gpuattest
HXC-1573	Feature	P2	Implement multi-GPU node enumeration in pkg/gpuattest
HXC-1579	Feature	P2	Implement pkg/security/admission OPA-style policy + signed-image (cosign) support
HXC-1580	Feature	P1	Implement pkg/modelintegrity hf_cache_verify gate (SHA-256 + size)
HXC-1581	Feature	P2	Implement model/revision anti-cheat verification token (V2 HMAC-SHA256)
HXC-1582	Feature	P3	Implement X25519 ephemeral session-key handshake for verification to model
HXC-1583	Feature	P1	Maintain pkg/scheduler cost-aware GPU placement plugin
HXC-1584	Feature	P1	Implement value-multiplier preemption in pkg/scheduler (Gepetto mode)
HXC-1585	Feature	P1	Implement SKIP LOCKED-style optimistic work-claiming in pkg/scheduler
HXC-1586	Feature	P1	Implement attestation-gated scheduler admission predicate
HXC-1587	Feature	P2	Implement bounty/auction economic-placement plugin for scheduler
HXC-1588	Feature	P2	Implement utilization-aware EWMA candidate ranking for scheduler/orchestrator
HXC-1589	Feature	P2	Implement Kueue-style suspend-then-admit tiered job admission
HXC-1590	Feature	P1	Implement NodeSelector placement-constraint schema in pkg/resources
HXC-1592	Feature	P1	Implement SUPPORTED_GPUS compute-multiplier catalog + LookupMux
HXC-1595	Feature	P1	Implement streaming-safe failover (retry only before first byte) in LLMOrchestrator
HXC-1596	Feature	P2	Implement per-task ordered fallback chains with dedup/cap + empty-responses
HXC-1597	Feature	P2	Implement thermal pre-warm / scale-from-zero state machine (Therm)
HXC-1598	Feature	P2	Implement passthrough cords + disconnect-aware upstream abort in LLMOrchestrator
HXC-1599	Feature	P3	Implement Claude/Responses API-shape adapters for LLMOrchestrator
HXC-1600	Feature	P3	Implement error-classified LLM failover taxonomy + sandbox scheduling
HXC-1601	Feature	P2	Implement pkg/inferenceproxy correlation-ID + backend audit trail
HXC-1602	Feature	P2	Implement deterministic keyed-hash anonymization in pkg/inferenceproxy
HXC-1603	Feature	P2	Implement spoof-proof managed-header sanitization at the recording edge
HXC-1604	Feature	P1	Maintain pkg/fiber length-prefixed framed miner<->validator transport
HXC-1605	Feature	P1	Implement fiber ed25519 signed-identity handshake + stake-gated admission
HXC-1606	Feature	P2	Implement node identity verify-then-pin (TOFU) keypair-rooted TLS
HXC-1607	Feature	P2	Implement pkg/marketplace registration -> bounty -> metering -> payout
HXC-1608	Feature	P2	Implement pkg/marketplace/audit commit-then-prove reproducible recording
HXC-1609	Feature	P3	Implement scale-to-zero metered hot/cold billing state machine
HXC-1610	Feature	P3	Implement watchtower-style liveness/integrity prober for served instances
HXC-1612	Feature	P3	Implement OpenAPI request-validation admission middleware for contracts
HXC-1613	Docs	P1	Document Phase 8C exit-gate evidence matrix (CLAUDE-1 usability pr

HXC	Type	Pri	Title
HXC-1614	Docs	P2	Document Phase 8C architecture integration diagrams (scheduler<-attese2ee<-TEE, marketplace)
HXC-1616	Bug	P2	Honor WithClock in cloudspot Azure/GCP pollers (currently silently ignore)
HXC-1621	Bug	P1	SELinux bind-mount relabel hardcoded :Z — make conditional + cross-arch (containers/pkg/crossbuild)
HXC-1626	Feature	P2	HXC-1119 follow-up: wire pkg/events Avro codec into the NATS Helix + schemas for SessionEvent/SchedulerEvent/AuditEvent
HXC-1627	Feature	P2	HXC-1626 follow-up: NATSBackend typed Avro publish/subscribe for SessionEvent/SchedulerEvent/AuditEvent
HXC-1628	Task	P2	buf lint STYLE conformance for api/v1 protos (PACKAGE_DIRECTORIES RPC-response naming)
HXC-1629	Task	P2	Wire Makefile migrate-up/migrate-down to scripts/run-migrations.sh (P1)
HXC-1630	Task	P1	Run govulncheck vulnerability scan across main/api/v1/security module (gap)
HXC-1631	Bug	P1	govulncheck: bump Go toolchain (go.mod go-directive) to clear 9 reachable advisories
HXC-1632	Bug	P2	govulncheck: bump golang.org/x/net v0.54.0 -> v0.55.0 (GO-2026-5026)
HXC-1633	Feature	P1	Implement real wireguard-go userspace path on macOS (CLAUDE-2 patch wireguard)
HXC-1634	Feature	P2	Implement real darwin GPU/device probes (Metal/IOKit) for internal/gpu (CLAUDE-2)
HXC-1635	Task	P2	Add SBOM generation + dependabot/renovate to complete supply-chain items 34/80)
HXC-1636	Task	P2	Local make deps-update target (no-CI dependency-update maintenance, P1)
HXC-1637	Bug	P1	make build is broken: builds non-existent ./cmd/helix-cluster
HXC-1638	Bug	P2	helixctl binary referenced in docs but absent from cmd/
HXC-1639	Bug	P2	SQL schema drift: 0001_primary_schema.sql diverges from the golang- (001-015)
HXC-1640	Feature	P1	pkg/stonith fencing-safety test (positive confirmation + fenced node cannot)
HXC-1641	Bug	P1	FIX registry NextHXCID lexical-vs-numeric id-allocation bug
HXC-1642	Feature	P1	pkg/epochresolve adversarial fencing test (stale epoch rejected, at-most-one)
HXC-1643	Bug	P0	FIX pkg/idempotent concurrency bug (data race + double-commit broke)
HXC-1644	Feature	P1	pkg/workclaim concurrent mutual-exclusion test (exactly-one claimant)
HXC-1645	Feature	P1	pkg/voting exhaustive at-most-one-authoritative test (split-brain invariant)
HXC-1646	Feature	P1	pkg/semaphore adversarial bounded-concurrency test (never exceed N, no)
HXC-1647	Feature	P1	pkg/lru adversarial eviction + capacity test (true LRU not FIFO, exact bound)
HXC-1648	Bug	P0	FIX pkg/lock double-release crash + cross-owner free (unguarded Unlock)
HXC-1649	Bug	P0	FIX pkg/heartbeatcoalescer Receiver data race (unsynchronized lastSeen)
HXC-1650	Bug	P1	harden EtdLocker fire-once release guard (follow-up to HXC-1648)
HXC-1651	Feature	P1	pkg/flowcontrol concurrent credit-conservation test (never over-admit)
HXC-1652	Feature	P1	pkg/ratelimit UserLimiter adversarial over-admission test (dual-bucket, P1)
HXC-1653	Feature	P1	

HXC	Type	Pri	Title
			pkg/pool GPU-instance-pool adversarial concurrency test (no double-iss bound)
HXC-1654	Feature	P1	pkg/watchmanager adversarial concurrency test (register/notify/sync chu
HXC-1655	Feature	P1	pkg/multiraft adversarial registry-concurrency test (concurrent create/ro
HXC-1656	Feature	P1	pkg/pubsub adversarial concurrent sub/unsub/publish test (no send-on-c
HXC-1657	Feature	P1	pkg/tieredcache coherence test (re-Put shadows stale lower tier, no conta
HXC-1658	Feature	P1	pkg/session adversarial concurrent FSM test (at-most-once terminal tran
HXC-1659	Bug	P1	FIX pkg/offlinesync Reconcile conflict winner non-deterministic (violat lower-Seq policy)
HXC-1660	Feature	P1	pkg/workqueue adversarial concurrency test (no double-processing, ded
HXC-1661	Feature	P1	pkg/metering conservation test (exact revenue accounting, per-key isolat
HXC-1662	Feature	P1	pkg/budgetcap concurrent over-spend test (cap never exceeded)
HXC-1663	Feature	P1	pkg/billingfsm adversarial money-safety test (transition matrix, no doub conservation)
HXC-1664	Feature	P1	pkg/informer cache-coherence test (latest-delta-wins, delete coherence, r
HXC-1665	Feature	P1	pkg/rebalance adversarial minimal-disruption test (sticky, balance, idem
HXC-1666	Bug	P0	FIX pkg/slotmigration concurrent-map crash (routing lookups raced the
HXC-1667	Bug	P0	FIX pkg/edgeregistry concurrent-map race (Register raced Get/Len/List
HXC-1668	Feature	P1	pkg/modelrouter concurrent read-safety test (immutable routing table)
HXC-1669	Feature	P1	pkg/fedtopology concurrent read-safety + path-correctness test (immutab
HXC-1670	Bug	P0	FIX pkg/jwt keystore concurrent-map crash on auth path (verify raced r
HXC-1671	Feature	P1	pkg/ewmarank concurrency-contract + ranking-correctness test (docume threaded)
HXC-1672	Bug	P0	FIX pkg/jobadmit concurrent-map crash (admit/complete raced state loo
HXC-1673	Feature	P1	pkg/failconfirm concurrency-contract + quorum-confirmation test (docu threaded)
HXC-1674	Bug	P1	FIX pkg/redundantexec concurrent-map + trust-field race (validate race
HXC-1675	Bug	P1	FIX pkg/rescorer concurrent ranking/map race (Tick rerank raced Top/R
HXC-1676	Feature	P1	pkg/fedtrust concurrent trust-policy test (immutable-after-build, deny-ne
HXC-1677	Bug	P1	FIX pkg/thermalwarm concurrent-map + state-field race (dispatch/state
HXC-1678	Feature	P1	pkg/phasegate concurrent read-safety + gating-correctness test (immutab
HXC-1679	Feature	P1	pkg/dataplane partial-lock audit + concurrent race test (Distributor.pendi
HXC-1680	Feature	P1	pkg/inferenceproxy partial-lock audit + concurrent race test (MemAudit
HXC-1681	Feature	P1	pkg/timefault concurrency-contract + fault-model test (deterministic sing harness)
HXC-1682	Feature	P1	pkg/auctionplace adversarial clearing test (first-caller-wins, order-indepe winner)
HXC-1683	Feature	P1	pkg/constraints Forbidden hard-ban precedence test (closes latent Forbid gap)
HXC-1684	Feature	P1	pkg/preempt 20k-case oracle preemption test (priority-safety, minimality
HXC-1685	Feature	P1	pkg/carbonsched oracle selection test (lowest-carbon argmin, order-inde

HXC	Type	Pri	Title
HXC-1686	Feature	P1	pkg/costsched oracle selection test (lowest-cost, lex tie-break; closes uns
HXC-1687	Feature	P1	pkg/suitability exhaustive placement test (HPC never remote/non-low-la
HXC-1688	Feature	P1	pkg/revenueopt adversarial pricing test (TEE->Chutes bonus, oracle win
HXC-1689	Feature	P1	pkg/forecast numeric OLS-correctness test (formula vs oracle, exact win
HXC-1690	Bug	P1	FIX pkg/qos intransitive near-tie comparator (order-dependent QoS sele
HXC-1691	Feature	P1	pkg/latencysched comparator transitivity audit + order-independence tes band)
HXC-1692	Feature	P1	pkg/gpupool comparator transitivity audit + order-independence test (tra
HXC-1693	Feature	P1	pkg/smartrouter dual audit (comparator transitivity + partial-lock) clean
HXC-1694	Feature	P1	pkg/bursthysteresis exhaustive hysteresis-boundary test (no flap, inclusi
HXC-1695	Feature	P1	pkg/healthmonitor FSM transition-matrix + concurrency test (edge-trigg guarded)
HXC-1696	Feature	P1	pkg/balancemonitor event-log conservation + concurrency test (no lost a threshold)
HXC-1697	Feature	P1	pkg/hybridkex adversarial hybrid-binding test (X25519+ML-KEM-768, bind)
HXC-1698	Feature	P1	pkg/x25519session adversarial key-agreement test (auth, wrong-peer, lo
HXC-1699	Feature	P1	pkg/modelintegrity adversarial tamper/forgery test (SHA-256 content-pi
HXC-1700	Feature	P1	pkg/gpuattest adversarial attestation-forgery test (ed25519, pinned-key t
HXC-1701	Feature	P1	pkg/spiffefed adversarial trust-domain-confusion test (exact host match, bypass)
HXC-1702	Task	P1	attestadmit fail-closed adversarial tests
HXC-1703	Task	P1	gravalverify anti-cheat adversarial tests
HXC-1704	Task	P1	exportcontrol embargo/lookalike adversarial tests
HXC-1705	Task	P1	imagepolicy admission adversarial tests
HXC-1706	Task	P1	headersanitize spoof-proofing adversarial tests
HXC-1707	Task	P1	edgeverify content-hash adversarial tests
HXC-1708	Task	P1	anonymize keyed-hash tokenizer adversarial tests
HXC-1709	Task	P1	priorityqueue total-order adversarial tests
HXC-1710	Task	P1	auditproof canonical-serialization adversarial tests
HXC-1711	Task	P1	backoff cap/overflow adversarial tests
HXC-1712	Task	P1	nodeselector constraint-satisfaction adversarial tests
HXC-1713	Task	P1	economics conservation/ROI adversarial tests
HXC-1714	Task	P1	hybridtco monotonicity/accounting adversarial tests
HXC-1715	Task	P1	costrouter cost-optimal selection adversarial tests
HXC-1716	Task	P1	fallbackchain orchestration adversarial tests
HXC-1717	Task	P1	openapivalidate fail-closed validation adversarial tests
HXC-1718	Task	P1	correlation audit-trail adversarial tests
HXC-1719	Task	P1	modelretry classification adversarial tests
HXC-1720	Task	P1	workloadrouter routing+concurrency adversarial tests
HXC-1721	Task	P1	streamfailover orchestration adversarial tests

HXC	Type	Pri	Title
HXC-1722	Task	P1	burstcapacity spillover-estimator adversarial tests
HXC-1723	Task	P1	powergater night-window/state adversarial tests
HXC-1724	Task	P1	residency data-residency compliance adversarial tests
HXC-1725	Task	P1	watchtower tamper-integrity adversarial tests
HXC-1726	Task	P1	fsresidency file content-integrity adversarial tests
HXC-1727	Task	P1	devicecatalog lookup adversarial tests
HXC-1728	Task	P1	gputopo topology-selection adversarial tests
HXC-1729	Task	P1	configsync CRDT store concurrency adversarial tests
HXC-1730	Task	P1	config defaults/validation adversarial tests
HXC-1731	Task	P1	deviceprofile YAML-load adversarial tests
HXC-1732	Task	P1	gpucatalog scoring/multiplier adversarial tests
HXC-1733	Task	P1	healthprobe grace-window adversarial tests
HXC-1734	Task	P1	devicemap round-trip mapping adversarial tests
HXC-1735	Task	P1	burst autoscaling-hysteresis adversarial tests
HXC-1736	Task	P1	archlint doc-vs-disk linter adversarial tests
HXC-1737	Task	P1	etcdlint import-ban linter adversarial tests
HXC-1738	Task	P1	tierdetect per-OS detection adversarial tests
HXC-1739	Task	P1	llmadapter translation-fidelity adversarial tests
HXC-1740	Task	P1	passthrough streaming-fidelity adversarial tests
HXC-1741	Task	P1	grpcutil auth-interceptor adversarial tests
HXC-1742	Task	P1	middleware panic-recovery/chain adversarial tests
HXC-1743	Task	P1	llmfailover failover-path/hint adversarial tests
HXC-1744	Task	P1	capability manifest/negotiate adversarial tests
HXC-1745	Task	P1	sandbox capability-isolation adversarial tests
HXC-1746	Task	P1	splitbrainalert detection adversarial tests
HXC-1747	Task	P1	serde capnp serialization adversarial tests
HXC-1748	Task	P1	netutil SSRF-defense adversarial tests
HXC-1749	Task	P1	healthmonitor FSM/concurrency adversarial tests
HXC-1750	Task	P1	leader fencing split-brain-safety adversarial tests
HXC-1751	Task	P1	workclaim claim-exclusivity adversarial tests
HXC-1752	Task	P1	tierdef config-load adversarial tests
HXC-1753	Bug	P0	hlc Update/Now logical-counter overflow breaks HLC domination (caus
HXC-1754	Task	P1	hashslot CRC16/hashtag adversarial tests
HXC-1755	Task	P1	mvcc snapshot-isolation adversarial tests
HXC-1756	Task	P1	stats Welch-t-test adversarial tests
HXC-1757	Task	P1	ringavg moving-average adversarial tests
HXC-1758	Task	P1	crdt semilattice-law adversarial tests
HXC-1759	Bug	P1	antientropy equal-version conflict never converges (permanent split-brain
HXC-1760	Task	P1	splitbrain quorum-classification adversarial tests

HXC	Type	Pri	Title
HXC-1761	Task	P1	porcupine linearizability-checker adversarial tests
HXC-1762	Bug	P1	offlinesync: decompression-bomb DoS + equal-Seq split-brain — BOTH
HXC-1763	Task	P1	heartbeatcoalescer coalescing/concurrency adversarial tests
HXC-1764	Task	P1	multiraft durability-safety adversarial tests
HXC-1765	Bug	P2	computeproto: FlatBuffers decode DoS (panic on hostile bytes) — hardened ReadComputeTaskSafe
HXC-1766	Task	P1	nattraversal STUN codec adversarial tests
HXC-1767	Task	P1	websocket envelope codec adversarial tests
HXC-1768	Task	P1	fiber framing/admission adversarial tests
HXC-1769	Task	P1	ice RFC8445 priority adversarial tests
HXC-1770	Task	P1	wasm sandbox-enforcement adversarial tests
HXC-1771	Bug	P1	helixnet prod fabric teardown race: send on closed channel panic — FIXED
HXC-1772	Task	P1	edgefusion windowed-fusion adversarial tests
HXC-1773	Task	P1	watchmanager watch-fanout concurrency adversarial tests
HXC-1774	Task	P1	informer cache-coherence adversarial tests
HXC-1775	Task	P1	cellmesh overlap-detection adversarial tests
HXC-1776	Task	P1	edgeheartbeat liveness/concurrency adversarial tests
HXC-1777	Task	P1	checkpoint_merge merge/migrate adversarial tests
HXC-1778	Task	P1	kraft metadata-log determinism adversarial tests
HXC-1779	Task	P1	gepetto reservation/high-water adversarial tests
HXC-1780	Task	P1	edgeregistry concurrency/copy-isolation adversarial tests
HXC-1781	Task	P1	marketplace/scan/chutes adversarial tests (over-commit, routing, attestations)
HXC-1782	Bug	P1	classads parser/evaluator stack-overflow DoS on deeply-nested expressions
HXC-1783	Task	P1	raftprofile parse/timing-invariant adversarial tests
HXC-1784	Task	P1	phasegate gating/concurrency adversarial tests
HXC-1785	Bug	P1	costtracker unsynchronized accumulator: data race + cost non-conservative
HXC-1786	Task	P1	providerchain fallback adversarial tests
HXC-1787	Task	P1	modelrouter selection adversarial tests
HXC-1788	Bug	P1	dataplane Distributor.SendToWorker concurrent-send libzmq SIGABRT
HXC-1789	Task	P1	marketplaceadapter translation-fidelity adversarial tests
HXC-1790	Task	P1	raftleader election-safety adversarial tests
HXC-1791	Bug	P1	budgetcap: NaN cost bypasses MaxCostPerHour cap (unbounded over-spend)
HXC-1792	Task	P3	metering: adversarial test pinning documented single-goroutine contract
HXC-1793	Task	P3	inferenceproxy: adversarial concurrency/edge hardening test
HXC-1794	Bug	P1	ewmarank: NaN sample poisons EWMA and breaks Rank total order (memory)
HXC-1795	Bug	P1	admissioncontrol: int64 reserve-sum overflow bypasses N+K capacity constraint
HXC-1796	Task	P3	auctionplace: adversarial test, FCFS board contract pinned (no bug)
HXC-1797	Task	P3	flowcontrol: adversarial test, APF round-robin cursor survives resort (no bug)
HXC-1798	Bug	P2	forecast: non-finite sample poisons OLS prediction and suppresses PreWelding scaling)

HXC	Type	Pri	Title
HXC-1799	Task	P3	gravaladmit: adversarial test, single-use nonce gate K=1 over-commit-pr
HXC-1800	Task	P3	balancemonitor: adversarial test, wire NaN/Inf rejected by stdlib JSON (
HXC-1801	Task	P3	fedtrust: adversarial test, deny branches fail-CLOSED (no bug)
HXC-1802	Bug	P2	carbonsched: NaN carbon-intensity poisons greenest-region selection +
HXC-1803	Task	P3	constraints: adversarial test, flat-struct placement engine (no bug)
HXC-1804	Task	P3	federation: adversarial test, uint32+enum total order, mutex-guarded (no
HXC-1805	Task	P3	discovery: adversarial test, all registries promise+keep concurrency safe
HXC-1806	Task	P3	cloudspot: adversarial test, preemption-drain handler deterministic (no b
HXC-1807	Task	P3	gpupool: adversarial test, stateless read-only query pkg (no bug)
HXC-1808	Task	P3	backfill: adversarial test, pure deterministic scheduler (no bug)
HXC-1809	Task	P3	health: adversarial test, fail-closed rollup, RWMutex-guarded (no bug)
HXC-1810	Bug	P2	rescorer: NaN price breaks rerank comparator -> NaN provider seizes to
HXC-1811	Task	P3	idempotent: adversarial test, exactly-once under concurrent first-use (no
HXC-1812	Task	P3	events: adversarial test, handler-func Bus race-free, no teardown-race su
HXC-1813	Task	P3	anticheat: adversarial test, HMAC token fail-CLOSED on all malformed
HXC-1814	Bug	P1	edge: Allowed() admits invalid/unknown TrustLevel to all workloads (fa
HXC-1815	Task	P3	failconfirm: adversarial test, strict-majority quorum + reporter dedup cor
HXC-1816	Task	P3	etcd: adversarial test, thin clientv3 wrapper + key-scheme pins (no bug)
HXC-1817	Task	P3	deviceplugin: adversarial test, Registry no-double-alloc + conservation (
HXC-1818	Bug	P2	gitops: PlanRollingSync silently drops apps with unknown/typo/empty t
HXC-1819	Bug	P2	hxcregistry: CreateItem/UpdateItem non-atomic -> orphan item rows wi
HXC-1820	Task	P3	fmea: adversarial test, int-only RPN scoring (no bug)
HXC-1821	Task	P3	gpu: adversarial test, provider-adapter registry RWMutex-guarded (no b
HXC-1822	Bug	P2	jobadmit: used+Request int overflow bypasses quota cap (over-admit +
HXC-1823	Bug	P2	smartrouter: NaN/Inf score latched at index 0 mis-routes (breaks all 4 str
HXC-1824	Task	P3	voting: adversarial test, strict-majority quorum + dedup + deterministic t
HXC-1825	Bug	P3	jwt: verifyRSA passes crypto.Hash(0) not SHA256/384/512 -> rejects al
HXC-1826	Bug	P1	security: IsValidTrustDomain admits '/' in SPIFFE trust domain (fail-op
HXC-1827	Bug	P2	storage: RedisStore.SetSessionRouting non-atomic -> orphan routing ha
HXC-1828	Bug	P2	validator: checkMin/checkMax float NaN bypasses range bounds (fail-o
HXC-1829	Task	P3	stonith: adversarial test, fencing confirms sink-side (no fail-open) (no bu
HXC-1830	Task	P3	metrics: adversarial test, atomic counters + mutex histograms conserve u
HXC-1831	Task	P3	session: adversarial test, Manager RWMutex-guarded + deep-copy Get -
HXC-1832	Bug	P2	retry: exponential/linear backoff overflows time.Duration -> negative de

HXC	Type	Pri	Title
HXC-1833	Task	P3	crypto: adversarial test, Ed25519/AES-GCM/PBKDF2 wrapper rejects t
HXC-1834	Task	P3	workerpool: adversarial test, bounded pool correct (no bug)
HXC-1835	Bug	P1	tiersec: tierRank Sscanf admits malformed tier names -> fail-open cleara
HXC-1836	Task	P3	tieredcache: adversarial test, LRU bound/eviction/recency correct (no bu
HXC-1837	Task	P3	doublecrypt: adversarial test, mTLS-over-WireGuard tamper/auth enforc
HXC-1838	Task	P3	covgate: adversarial test, threshold >= boundary + NaN fail-closed (no b
HXC-1839	Bug	P1	qualitygate: NaN metric/threshold records no breach -> CRITICAL gate poisoned metric (fail-open, CLAUDE-1)
HXC-1840	Bug	P2	inference: 2 stale tests fail on main (encode retired internal/llm synthetic RED
HXC-1841	Task	P3	pool: adversarial test, mutex-guarded pool no over-commit/double-hand
HXC-1842	Task	P3	quantization: adversarial test, int-only VRAM estimate (no bug)
HXC-1843	Task	P3	raft: adversarial test, hashicorp/raft wrapper + local FSM safety pinned (
HXC-1844	Task	P3	device: adversarial test, deterministic classifier + real per-OS probe, CL (no bug)
HXC-1845	Task	P3	dst: adversarial test, deterministic-simulation harness is genuinely deter
HXC-1846	Task	P3	provider: adversarial test, pure HTTP adapter no selector/registry (no bu
HXC-1847	Bug	P2	timefault: abs64(MinInt64) overflow misses max-skew detection -> gran clock (split-brain)
HXC-1848	Task	P3	devicemap: adversarial concurrency test, pure stateless conversion (no b
HXC-1849	Task	P3	fedtopology: adversarial test, deterministic immutable topology (no bug)
HXC-1850	Task	P3	thermalwarm: adversarial test, duration-based warm-up boundary correc
HXC-1851	Bug	P2	tracing: Span.Duration() ignores endTime -> finished span duration grov (wrong latency)
HXC-1852	Bug	P2	agentprovision: ModeManager.Submit no duplicate-ID guard -> double- live-job eviction
HXC-1853	Task	P3	chaosexp: adversarial test, conservation + determinism (no bug)
HXC-1854	Task	P3	errors: adversarial test, Is/As wrap-chain + Wrap(nil)==nil (no bug)
HXC-1855	Bug	P1	swim: handleDead lacks self-target guard -> node accepts death rumour denial of liveness)
HXC-1856	Bug	P2	build: Job.clone() aliases BuildArgs map despite documented deep-copy mutation corrupts canonical job
HXC-1857	Task	P3	log: adversarial test, slog shared-mutex concurrent integrity (no bug)
HXC-1858	Task	P3	bootstrap: adversarial test, peer-rendezvous deterministic + no fail-open
HXC-1859	Task	P3	wireguard: adversarial test, CLAUDE-2 darwin parity REAL (wireguard race-clean (no bug)
HXC-1860	Task	P3	grafanadash: adversarial test, deterministic JSON + escaping safe (no bu
HXC-1861	Task	P3	helixtask: adversarial test, pure stateless decorator (no bug)
HXC-1862	Task	P3	chutesaccount: adversarial test, read-only client no money-mutation sur
HXC-1863	Bug	P2	scheduler: Schedule/ScheduleOptimistic rank over map iteration -> non placement on score ties
HXC-1864	Bug	P2	

HXC	Type	Pri	Title
			infra: containerOrchestrator.Scale no negative-replicas guard -> slice-bo (process crash)
HXC-1865	Bug	P2	local: GPU.validate() lets NaN/Inf slip ordered guards -> poisons TCO, forever
HXC-1866	Task	P3	phase7matrix: adversarial test, recompute + boundary correct, fail-closed
HXC-1867	Bug	P2	context: Detach leaks parent Deadline() despite 'never cancelled' -> self open
HXC-1868	Task	P3	benchmark: adversarial test, CoefficientOfVariation Bessel/zero-mean co
HXC-1869	Task	P3	compliancedoc: adversarial test, deterministic model card, no fabricated
HXC-1870	Task	P3	e2eebench: adversarial test, nearest-rank percentile + ML-KEM key-agr bug)
HXC-1871	Task	P3	resources: adversarial test, CLAUDE-2 darwin parity VERIFIED REAL completes pkg/ 196/196
HXC-1872	Bug	P1	internal/node: UpdateNodeStatus mutates shared registry *Node pointer race (live in gRPC server)
HXC-1873	Task	P3	internal/gpu: adversarial test, GPU allocator no over-commit + CLAUD (no bug)
HXC-1874	Task	P3	internal/gateway: adversarial test, authorize fail-CLOSED (no bug)
HXC-1875	Task	P3	internal/security: adversarial test, identity/authz/policy all fail-CLOSED
HXC-1876	Bug	P1	internal/policy: scheduling rego permits non-numeric node.health -> ren (schedules unhealthy node)
HXC-1877	Bug	P2	internal/messaging: Queue.Enqueue releases lock before send -> send-on racing Close (teardown race)
HXC-1878	Task	P3	internal/scheduler: adversarial test, gRPC ScheduleJob serialized no ove
HXC-1879	Task	P3	internal/llm: adversarial test, Manager RWMutex-guarded + fail-closed
HXC-1880	Bug	P2	internal/build: SubmitBuild stores spec.BuildArgs by reference -> caller worker read (shared-pointer alias)
HXC-1881	Bug	P2	internal/federation: scoreCell NaN score breaks cell-selection total order admitted as Chosen
HXC-1882	Bug	P1	internal/health: aggregate switch has no default -> malformed/unknown HEALTHY (fail-open readiness/liveness)
HXC-1883	Task	P3	internal/console: adversarial test, Registrar stateless no shared-pointer ra
HXC-902	Task	P1	
HXC-903	Task	P1	
HXC-904	Feature	P0	Phase 4 Build Service
HXC-905	Feature	P0	SQLite HXC Registry
HXC-908	Bug	P1	Harden pkg/stonith IPMI credential handling (no -P on argv, redact in en
HXC-909	Task	P2	pkg/multiraft: make RaftTransport async-delivery safe (Step under share
HXC-910	Task	P2	pkg/deviceplugin: add concurrency test exercising Registry mutex under
HXC-911	Bug	P2	pkg/chutes StreamChannel: honor cancellation on blocked reads + close
HXC-912	Bug	P2	pkg/kraft: CreateTopic should return ErrTopicExists on conflicting re-cr
HXC-913	Task	P3	pkg/porcupine: implement state-dedup memoization via Model.Equal fo

HXC	Type	Pri	Title
HXC-915	Task	P2	pkg/dst: wire 1000-sims/commit deterministic-simulation CI gate (GH A harness)
HXC-916	Task	P2	pkg/provider/chutes: ctx-interruptible backoff + honor Retry-After header
HXC-917	Bug	P2	pkg/multiraft: handle ShardStorage SetHardState/Append errors before
HXC-918	Bug	P2	pkg/marketplaceadapter Akash: default MinReputation via constructor +
HXC-919	Bug	P2	pkg/provider/runpod: surface Worker.Endpoint + don't hold lock across network call
HXC-920	Task	P3	pkg/provider/chutes: test parseRetryAfter HTTP-date branch (RFC1123 + past-date clamp)
HXC-927	Task	P2	Surface multiraft durability persist-error through public API
HXC-934	Bug	P0	Make cmd/e2ee-proxy consume security/pkg/e2ee instead of inline crypto
HXC-937	Task	P1	Add e2e + chaos test types for consensus/federation/marketplace
HXC-938	Task	P2	Add fuzz to crypto and benchmarks to hot paths
HXC-939	Docs	P2	Refresh stale per-phase GAP_AUDIT.md files (CLAUDE-3)
HXC-940	Task	P2	Wire the orphaned gate packages (covgate/archlint/etcdlint/qualitygate/p
HXC-941	Task	P2	Reconcile E2EE AEAD spec deviation (AES-256-GCM vs roadmap Cha
HXC-942	Bug	P1	Add e2e/chaos -race integration tests for the 8 fixed concurrency hazard
HXC-943	Bug	P1	Test HelixQA validators (done) and vision/ORB (blocked) + fix helixqa
HXC-944	Bug	P0	Fix 3 CRITICAL CVEs from trivy security scan
HXC-945	Bug	P1	Triage and resolve gosec HIGH findings (99 HIGH) from security scan
HXC-946	Task	P2	Triage trivy misconfig HIGH (50) + suppress QA-sentinel secret false-p
HXC-947	Task	P3	Add root go.mod require/replace for digital.vasic.security (out-of-works
HXC-948	Bug	P2	Fix api/v1 module path /v1-suffix blocking out-of-workspace require/rep
HXC-950	Bug	P1	sidecarutil HealthProbe load-robustness (WaitDelay + transient retry)
HXC-951	Research	P2	TLA+ spec for SWIM membership/failure-detection (TLC-verified safer
HXC-952	Feature	P2	Unified hardware-inventory engine (CPU+memory+GPU+NPU aggrega
HXC-953	Feature	P2	Real Raft-backed leader-election service (pkg/raftleader adapting pkg/ra
HXC-954	Feature	P1	Edge gateway multi-transport integration (QUIC + MQTT + WebSocket
HXC-955	Feature	P2	Distributed KV demo over Raft (cmd/raftkv-demo) — building blocks c
HXC-956	Feature	P2	OpenTelemetry single-runtime tracing foundation (observability/tracing)
HXC-957	Research	P2	TLA+ spec for WireGuard/Noise-IK handshake (TLC-verified safety)
HXC-958	Research	P2	TLA+ spec for OR-Set CRDT convergence (TLC-verified, add-wins)
HXC-959	Research	P2	TLA+ spec for Two-Phase Commit atomicity (TLC-verified)
HXC-960	Research	P2	TLA+ spec for Saga compensation atomicity (TLC-verified)
HXC-961	Research	P2	TLA+ spec for vector-clock causal delivery (TLC-verified)
HXC-962	Research	P2	TLA+ spec for Raft single-server membership change safety (TLC-verifi
HXC-963	Research	P2	TLA+ spec for leader-lease leaseholder reads / no-stale-read (TLC-verif
HXC-964	Research	P2	TLA+ spec for phi-accrual failure-detector accuracy (TLC-verified)
HXC-965	Feature	P1	End-to-end cluster node-agent: compose discovery+raft+capabilities+ga (clusternode)

HXC	Type	Pri	Title
HXC-966	Research	P2	TLA+ spec for Chandy-Lamport distributed snapshot (TLC-verified)
HXC-967	Research	P2	TLA+ spec for Lamport distributed mutual exclusion (TLC-verified)
HXC-968	Research	P2	TLA+ spec for Dynamo-style quorum consistency $R+W>N$ (TLC-verified)
HXC-969	Feature	P1	clusternode real-WebSocket transport: cross-node messaging over real TCP
HXC-970	Research	P2	TLA+ spec for Hybrid Logical Clock (TLC-verified)
HXC-971	Research	P2	TLA+ spec for Raft ReadIndex linearizable reads (TLC-verified)
HXC-972	Research	P2	TLA+ spec for STONITH fencing safety / no dual-active (TLC-verified)
HXC-973	Research	P2	TLA+ spec for gossip/epidemic dissemination convergence (TLC-verified)
HXC-974	Feature	P1	Real TCP-networked Raft for pkg/raft (network transport + leak-safe cluster)
HXC-975	Research	P2	TLA+ spec for PN-Counter CRDT convergence (TLC-verified)
HXC-976	Feature	P1	Real BoltDB persistence for pkg/raft (durable on-disk store + recovery)
HXC-977	Research	P2	TLA+ spec for rendezvous (HRW) consistent hashing minimal disruption
HXC-978	Feature	P1	pkg/raft: combined persistent+networked node (real TCP transport + on-disk store + restart-recover-rejoin)
HXC-979	Research	P1	TLA+ snapshot-install / log-compaction safety spec (RaftSnapshot), TLC-verified exhaustive
HXC-980	Bug	P1	registry CLI: update auto-moves current_location on status change + record (doc-sync truthfulness)
HXC-981	Feature	P1	pkg/wasm: combined-bounds sandbox E2E (fuel + memory cap + capabilities simultaneously)
HXC-982	Feature	P0	helix-raftd: real multi-process persistent Raft daemon (TCP + BoltDB) with from-disk E2E
HXC-983	Docs	P1	docs/consensus.md: consolidated Raft consensus architecture doc + READER docs_chain tracking
HXC-984	Feature	P1	pkg/raft: snapshot/log-compaction InstallSnapshot integration test (runtime pairing of HXC-979)
HXC-985	Feature	P1	clusternode durable consensus: NodeAgent on persistent+networked raft with from-disk E2E
HXC-986	Research	P1	TLA+ Raft PreVote spec — stale/restarted node cannot disrupt a stable leader (TLC-verified)
HXC-987	Feature	P1	helix-raftd Prometheus /metrics endpoint (live raft state, real scrape proven)
HXC-988	Feature	P1	e2ee adversarial tamper/wrong-key crypto tests (AEAD tag, ML-KEM-768 rejection)
HXC-989	Docs	P2	helix-raftd observability materials: Prometheus scrape config + Grafana dashboard observability.md
HXC-990	Feature	P1	pkg/raft PreVote runtime disruption test (runtime pairing of HXC-986)
HXC-991	Feature	P2	helix-raftctl CLI client for helix-raftd admin API (live-cluster proven)
HXC-992	Feature	P1	pkg/scheduler Omega optimistic-concurrency barrier-race test (no double-count)
HXC-993	Task	P2	gofmt hygiene: format 125 unformatted committed Go files across pkg/
HXC-994	Bug	P0	FIX real SWIM false-positive failure detection (healthy members wrong)
HXC-995	Feature	P1	pkg/antientropy end-to-end Merkle reconciliation test (only-differences-allowed)
HXC-996	Feature	P1	pkg/mvcc snapshot-isolation + race-safety test (consistent point-in-time)

HXC	Type	Pri	Title
HXC-997	Feature	P1	pkg/deltacrdt adversarial convergence tests + characterizes a real ORSet
HXC-999	Feature	P1	pkg/splitbrain partition-safety test (at-most-one-authoritative across all c